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RETENTION TIME

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1 ABSTRACT

1.1 Experimental purpose

- Survey retention time of a system of continuous stirred-tank reactors (CSTRs).
- Determine real retention time distribution function and compare with theoretical retention time distribution function.

1.2 Experimental results

- Maximum absorbance (for 1 reactor): $D_0 = 0.03$
- Reactor diameter: $d = 14.8$ (cm)
- Water level in the CSTR: $H = 12$ (cm)
- Flowrate of water: $v = 0.4$ (liter/min)

2 EXPERIMENTAL THEORIES

2.1 Basic concept

In the equipment system, different fluid elements will follow different paths. Based on the defined retention time distribution function, we can evaluate the flows in the equipment, identify drawbacks when designing (such as dead zones), and find ways to overcome them.

The retention time of an element in the system is the time that the element stays in the reactor. Retention time is a probability quantity. All retention times are around the average retention time; therefore, determining the average retention time is particularly significant.

$$\bar{t} = \frac{\sum t_{Vi}}{n} \quad (1)$$

Where t_{Vi} is the retention time of element i .

With the definition of retention time distribution function $F(t_V) = E(t_V)$:

$$E(t) = \frac{C(t)}{\int_0^\infty C(t)dt} \quad (2)$$

Average retention time according to volume:

$$\tau = \frac{V_R}{v} \quad (3)$$

Where:

- V_R : Volume of fluid in the reactor (liters).

- v : Flowrate of the input (liters/sec).

If the indicator does not reach the ideal correlation, the above equation is not satisfied. If $\bar{t} \neq \tau$, the indicator may be absorbed into the wall or there may be dead zones.

2.2 Methods of studying retention time

The retention time study can be carried out according to the following methods:

1. Determine the composition of components at time t of leaving the equipment, determine function $F(t)$.
2. Determine the composition of components at time t of still remaining in the equipment, function $I(t)$.
3. Determine the composition of components at time t of being out of the equipment, function $E(t)$.

To survey the performance of a real reactor, we often use the stimulus-response method (marking method).

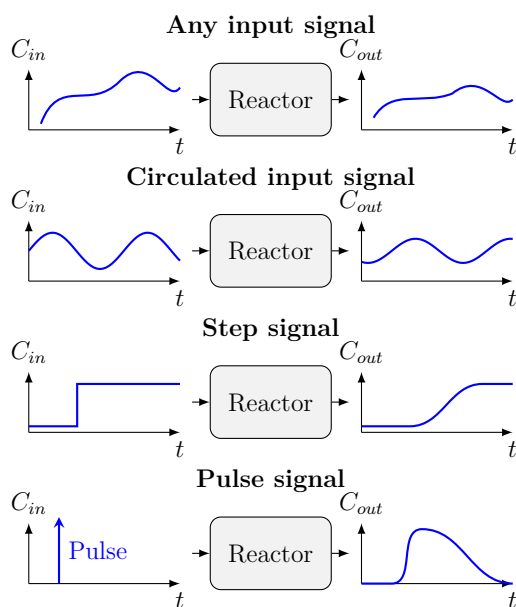


Figure 1: Frequently-used stimulus-response signals

In this experiment, we use the pulse signal (Dirac delta function).

2.3 Retention time distribution function of CSTRs model

The general form of the distribution function for n reactors in series:

$$E(\theta) = \frac{n(n\theta)^{n-1}}{(n-1)!} e^{-n\theta} \quad (4)$$

When $n = 1$, we have an ideal CSTR model. When $n = \infty$, we have an ideal plug flow model.

Dimensionless compact time:

$$\theta = \frac{t}{\tau} \quad (5)$$

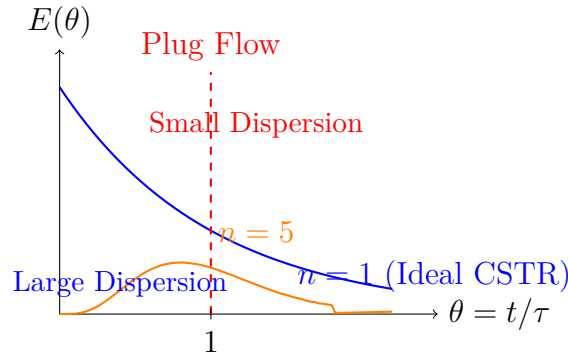


Figure 2: The curve C (E-curve) showing the response at output flow for the pulse signal at the input flow for different tanks in series

3 EXPERIMENTAL EQUIPMENT

The following equipment is used in the experiment:

- Continuous stirred-tank reactors (CSTRs) system
- Test tubes
- Stopwatch
- Pipette
- Colored substance (indicator)
- Light transmittance (absorbance) meter (Photometer)

4 EXPERIMENTAL METHODS

Proceed in the following order:

1. Open the valve for water to flow into the high-positioned tank until there is water in the overflowing tube.
2. Open the valve for water to flow through the flow meter into the CSTRs system and adjust the flowrate.
3. **System of one reactor:** When the system is stable, put the colored substance into reactor 1. Start the stopwatch at the same time. Use test tubes to take samples from time to time, then measure the absorbance.

4. **System of two and three reactors:** Do the same as for one reactor; put the colored substance into reactor 1 and take samples at the last reactor (from the last interconnected pipe).

5 EXPERIMENT RESULTS

5.1 Raw results

Table 1: Raw results

t (min)	1 tank system	2 tank system	3 tank system
1	0.013	0.005	0.027
2	0.026	0.025	0.02
3	0.013	0.018	0.014
4	0.011	0.008	0.013
5	0.014	0.013	0.016
6	0.003	0.004	0.001
7	0.002	0.011	0.002
8	0.019	0.019	0.005
9	0.006	0.002	0.003
10	0.026	0.002	0.005
11	0.009	0.023	0.012
12	0.004	0.001	0.011
13	0.018	0.001	0.003
14	0.011	0.005	0.002
15	0.002	0.006	0.01
16	-	0.003	0.003
17	-	0.004	0.005
18	-	0.001	0.004
19	-	0.004	0.004
20	-	0.001	0.009
21	-	0.009	0.002
22	-	0.002	0.002
23	-	0.001	0.005
24	-	0.02	0.008
25	-	0.022	0.008
26	-	-	0.004
27	-	-	0.01
28	-	-	0.009
29	-	-	0.01
30	-	-	0.004
31	-	-	0.005
32	-	-	0.003
33	-	-	0.005
34	-	-	0.004
35	-	-	0.01

5.2 1 tank system

Table 2: 1 tank system results

t (min)	D	t.D	θ (Exp)	E (Exp)	θ (Theo)	E (Theo)
1	0.013	0.013	0.14	0.43	0.19	0.82
2	0.026	0.052	0.28	0.87	0.39	0.68
3	0.013	0.039	0.41	0.43	0.58	0.56
4	0.011	0.044	0.55	0.37	0.78	0.46
5	0.014	0.070	0.69	0.47	0.97	0.38
6	0.003	0.018	0.83	0.10	1.16	0.31
7	0.002	0.014	0.97	0.07	1.36	0.26
8	0.019	0.152	1.11	0.63	1.55	0.21
9	0.006	0.054	1.24	0.20	1.74	0.17
10	0.026	0.260	1.38	0.87	1.94	0.14
11	0.009	0.099	1.52	0.30	2.13	0.12
12	0.004	0.048	1.66	0.13	2.33	0.10
13	0.018	0.234	1.80	0.60	2.52	0.08
14	0.011	0.154	1.93	0.37	2.71	0.07
15	0.002	0.030	2.07	0.07	2.91	0.05
Total	0.177	1.281				

The values needed to be calculated:

- $D_0 = 0.03$ (m)
- $t_{tb} = 7.24$ (min)
- $v = 0.0004$ (m^3/min)
- $V = 0.0021$ (m^3)
- $\tau = 5.16$ (min)

Placeholder: Relation $D/D_0 = f(\theta)$ for 1-tank system

Figure 3: The graph shows the relation $D/D_0 = f(\theta)$ for the theoretical and empirical case of a 1-tank system

5.3 2 tank system

Table 3: 2 tank system results

t (min)	D	t.D	θ (Exp)	E (Exp)	θ (Theo)	E (Theo)
1	0.005	0.005	0.09	0.17	0.10	0.32
2	0.025	0.05	0.17	0.83	0.19	0.53
3	0.018	0.054	0.26	0.60	0.29	0.65
4	0.008	0.032	0.34	0.27	0.39	0.71
5	0.013	0.065	0.43	0.43	0.48	0.74
6	0.004	0.024	0.51	0.13	0.58	0.73
7	0.011	0.077	0.60	0.37	0.68	0.70
8	0.019	0.152	0.69	0.63	0.78	0.66
9	0.002	0.018	0.77	0.07	0.87	0.61
10	0.002	0.02	0.86	0.07	0.97	0.56
11	0.023	0.253	0.94	0.77	1.07	0.51
12	0.001	0.012	1.03	0.03	1.16	0.45
13	0.001	0.013	1.11	0.03	1.26	0.41
14	0.005	0.07	1.20	0.17	1.36	0.36
15	0.006	0.09	1.29	0.20	1.45	0.32
16	0.003	0.048	1.37	0.10	1.55	0.28
17	0.004	0.068	1.46	0.13	1.65	0.24
18	0.001	0.018	1.54	0.03	1.74	0.21
19	0.004	0.076	1.63	0.13	1.84	0.19
20	0.001	0.02	1.71	0.03	1.94	0.16
21	0.009	0.189	1.80	0.30	2.03	0.14
22	0.002	0.044	1.88	0.07	2.13	0.13
23	0.001	0.023	1.97	0.03	2.23	0.10
24	0.02	0.48	2.06	0.67	2.33	0.09
25	0.022	0.55	2.14	0.73	2.42	0.08
Total	0.21	2.451				

The values needed to be calculated:

- $D_0 = 0.015$ (m)
- $t_{tb} = 11.67$ (min)
- $v = 0.0004$ (m^3/min)
- $V = 0.0041$ (m^3)
- $\tau = 10.32$ (min)

Placeholder: Relation $D/D_0 = f(\theta)$ for 2-tank system

Figure 4: The graph shows the relation $D/D_0 = f(\theta)$ for the theoretical and empirical case of a 2-tank system

5.4 3 tank system

Table 4: 3 tank system results

t (min)	D	t.D	θ (Exp)	E (Exp)	θ (Theo)	E (Theo)
1	0.027	0.027	0.07	0.90	0.06	0.03
2	0.02	0.04	0.13	0.67	0.13	0.10
3	0.014	0.042	0.20	0.47	0.19	0.20
4	0.013	0.052	0.26	0.43	0.26	0.32
5	0.016	0.08	0.33	0.53	0.32	0.43
6	0.001	0.006	0.40	0.03	0.39	0.54
7	0.002	0.014	0.46	0.07	0.45	0.64
8	0.005	0.04	0.53	0.17	0.52	0.73
9	0.003	0.027	0.60	0.10	0.58	0.81
10	0.005	0.05	0.66	0.17	0.65	0.87
11	0.012	0.132	0.73	0.40	0.71	0.92
12	0.011	0.132	0.80	0.37	0.78	0.95
13	0.003	0.039	0.86	0.10	0.84	0.98
14	0.002	0.028	0.93	0.07	0.90	0.99
15	0.01	0.15	1.00	0.33	0.97	1.00
16	0.003	0.048	1.06	0.10	1.03	0.99
17	0.005	0.085	1.13	0.17	1.10	0.98
18	0.004	0.072	1.20	0.13	1.16	0.96
19	0.004	0.076	1.26	0.13	1.23	0.94
20	0.009	0.18	1.33	0.30	1.29	0.91
21	0.002	0.042	1.40	0.07	1.36	0.88
22	0.002	0.044	1.46	0.07	1.42	0.84
23	0.005	0.115	1.53	0.17	1.49	0.81
24	0.008	0.192	1.60	0.27	1.55	0.77
25	0.008	0.2	1.66	0.27	1.61	0.74
26	0.004	0.104	1.73	0.13	1.68	0.70
27	0.01	0.27	1.80	0.33	1.74	0.67
28	0.009	0.252	1.86	0.30	1.81	0.63
29	0.01	0.29	1.93	0.33	1.87	0.60
30	0.004	0.12	2.00	0.13	1.94	0.57
31	0.005	0.155	2.06	0.17	2.00	0.53
32	0.003	0.096	2.13	0.10	2.07	0.50
33	0.005	0.165	2.20	0.17	2.13	0.47
34	0.004	0.136	2.26	0.13	2.20	0.45
35	0.01	0.35	2.33	0.33	2.26	0.42
Total	0.226	3.407				

The values needed to be calculated:

- $D_0 = 0.01$ (m)
- $t_{tb} = 15.08$ (min)

- $v = 0.0004 \text{ (m}^3/\text{min)}$
- $V = 0.0062 \text{ (m}^3)$
- $\tau = 15.48 \text{ (min)}$

Placeholder: Relation $D/D_0 = f(\theta)$ for 3-tank system

Figure 5: The graph shows the relation $D/D_0 = f(\theta)$ for the theoretical and empirical case of a 3-tank system

6 DISCUSSION

6.1 Compare the retention time distribution function of theory and experiment

- **1 tank system:** As seen in Figure 3, the retention time distribution function of the experiment is totally different compared with the theory. Generally, the E values of the theory are smaller than the experiment. Besides, the compact time values (θ) of theory are larger than the experiment ones. This suggests that according to the theory, dead zones or bypasses might appear. However, since the tank has a simple design, bypasses are unlikely; the discrepancy is likely due to dead zones where the agitators cannot reach.
- **2 tank system:** Following Figure 4, the retention time distribution function of the experiment somewhat resembles the theory (the E value increases to a maximum and then decreases). However, the increase is not regular and is quite fluctuated. Additionally, the E values of the theory are higher than the experiment.
- **3 tank system:** The retention time distribution function of the experiment is completely different compared with the theory and is extremely fluctuated.

6.2 Phenomena causing unsteady state in process and equipment

- **Dead zones:** Created because the speed of the agitators is too slow, leaving some volumes unstirred.
- **Fluid circulation:** Caused by uneven mixing due to fluid properties or slow agitator speed.
- **Overcoming phenomena:** These can be mitigated by increasing the agitator speed for more uniform mixing.
- **Other factors:**
 - Alteration of speed and movement direction due to equipment geometry (reactor, agitators, piping), creating swirl or dead areas.

- Unstable rotor speed over time (due to motor or voltage fluctuations).
- Indicator absorption onto the equipment walls.
- Ambient temperature fluctuations altering fluid properties (viscosity, density, velocity).

6.3 Evaluation of Errors

6.3.1 Error due to equipment

- **Photometer:** This is the primary source of error.
 - Dirty cuvettes or samples absorb light, influencing results.
 - The photometer is highly sensitive; even minor smudges on the cuvette surface affect readings.
 - Aging of the machine and unstable power supply.
- **Agitators:** Uneven swirling and unsteady velocity due to voltage fluctuations.
- **Piping:** Resistance in valves and piping makes it difficult to maintain constant flowrates, especially in multi-tank systems. Volume instability in the tanks during the experiment affects the steady state.

6.3.2 Operation errors

- Timing precision when taking samples.
- Improper cleaning or drying of cuvettes, leading to contamination between samples.
- Unstable control of flowrates.

6.3.3 Errors because of initial moment and pulse signal

It is difficult to establish the exact Dirac delta function because the injection of the indicator takes a finite amount of time (approx. 5 seconds), which is significant and causes discrepancies between the theoretical and experimental curves.

7 CALCULATION

7.1 Calculated values for 1 tank system

7.1.1 Experiment

Maximum absorption: $D_0 = 0.03$

Average retention time:

$$\bar{t} = \frac{\sum D_i t_i}{\sum D_i} \quad (6)$$

Compact time:

$$\theta = \frac{t}{\tau} \quad (7)$$

Measurement of retention time's distribution:

$$E_\theta = \frac{D}{D_0} \quad (8)$$

7.1.2 Theory

Water flowrate input to the tank: $v = 0.4$ (l/min) $= 0.4 \times 10^{-3}$ (m^3/min)

Tank volume: $V = \frac{\pi}{4} \times 0.148^2 \times 0.12 = 0.0021$ (m^3)

Volume time:

$$\tau = \frac{V}{v} = 5.16 \text{ min} \quad (9)$$

Compact time:

$$\theta = \frac{t}{\tau} \quad (10)$$

Measurement of retention time's distribution:

$$E_\theta = e^{-\theta} \quad (11)$$

7.2 Calculated values for 2 tank system

7.2.1 Experiment

Maximum absorption: $D_0 = 0.03/2 = 0.015$

7.2.2 Theory

Tank volume: $V = 2 \times \frac{\pi}{4} \times 0.148^2 \times 0.12 = 0.0041$ (m^3)

Volume time:

$$\tau = \frac{V}{v} = 10.32 \text{ min} \quad (12)$$

Measurement of retention time's distribution ($N = 2$):

$$E_\theta = \frac{N(N\theta)^{N-1}}{(N-1)!} e^{-N\theta} = 2(2\theta)e^{-2\theta} \quad (13)$$

7.3 Calculated values for 3 tank system

7.3.1 Experiment

Maximum absorption: $D_0 = 0.03/3 = 0.01$

7.3.2 Theory

Tank volume: $V = 3 \times \frac{\pi}{4} \times 0.148^2 \times 0.12 = 0.0062 \text{ (m}^3\text{)}$

Volume time:

$$\tau = \frac{V}{v} = 15.48 \text{ min} \quad (14)$$

Measurement of retention time's distribution ($N = 3$):

$$E_\theta = \frac{N(N\theta)^{N-1}}{(N-1)!} e^{-N\theta} = \frac{3(3\theta)^2}{2} e^{-3\theta} \quad (15)$$

8 REFERENCES

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